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CART 470 Capstone

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Journal 6th

During the week after our meeting with Jonathan, where we presented our three ideas and how it would look like and gameplay prototypes, He liked the jenga one and later on the maze mosaic one but wasn't very fond of the soccer game. So we ruled it out, for the jenga one, he liked it but he was concerned with latency while for the mosaic, we explained more about mixing both maze, and the mosaic idea together. He had mentioned about not liking mazes but we delved into more ideas in depth of it and different ways to do it aside from the normal maze layout. Afterwards, we started preparing documents with the information of what we had done so far and tried to narrow all that information down and looked over if there were things needed to be added before finishing the powerpoint. We had to fix it up as it was too wordy, so we limited the amount of words and brought a lot of our work reference into the powerpoint.

CART 470

1. Who is your client?

Client: Jonathan Lessard (CART)

Jonathan Lessard wants to create a **large-crowd multiplayer game** for the CART 415 Winter studio, to be presented at the **end-of-year exhibition**.

Visitors will **scan a QR code to connect their phones** and **control** the avatars on a shared projected screen.

Key point: Our client is interested in both the game idea and the practical process of developing it with a large team within a limited production timeline.

2. Project Background / Context

Our role: Research, develop, and prototype game concepts to identify potential challenges and provide a strong foundation for next semester's students.

- The project explores how a large team can collaboratively develop a simple, persistent, asynchronous multiplayer game.
- The game will be presented in an exhibition environment using a projector or large screen.
- Visitors will scan a QR code to access a webpage on their phones, which will become their controllers. Their characters will then appear on the projected screen.
- The experience should support approximately 20–30 players while allowing visitors to join and leave at any time. But also be playable with just 1 person
- The game should remain active without relying on a host.
- Our task is to explore the game mechanics, visual design, rules, scoring, and technical challenges before handing the project over to next semester's students.
- He wants us to explore possible game concepts, identify risks early, and establish a clear direction before the next group of CART 415 students begins development.
- Rather than delivering a finished game, our responsibility is to create a prototype and foundation the next class can realistically build on.

Main design requirements:

- Simple and immediately understandable gameplay.
- Persistent play that accommodates players joining and leaving.
- Multiplayer interactions that remain manageable with many participants.

- A clear objective and visible feedback for player actions.
- A leaderboard that communicates scores and progress.
- A feasible scope for the next class to implement.

3. Initial Research

Our initial research focused on identifying suitable game structures and understanding what makes an asynchronous, exhibition-based multiplayer experience work.

- Game references: We discussed games such as Baba Is You and Sokoban as references for clear rules, spatial problem-solving, and object interactions.
- We looked up any multiplayer games where you connect by phone:
- Game objectives and feedback: We asked what players are trying to achieve, how they know they have succeeded, and how their actions contribute to a score or shared objective.
- Visual communication: We considered communicating the rules directly through the projected screen rather than using a large menu or lengthy tutorial.
- Technical feasibility: We discussed the need to identify potential problems early so the next group can focus on implementation instead of redesigning the entire game.

Our early brainstorming produced several possible mechanics, including collecting objects, colouring tiles, matching a design, pushing blocks, building a tower, and competing in a soccer game.

The client encouraged us to explore different approaches

4. Project Development Process

We had ideation and the Prototype

Ideation:

- The first 2 or 3 weeks were ideation
- We each came up with multiple ideas, narrowed down to 3
- After we all looked at each game through a different de-risking lens
- Then we made different types of prototypes for each; then we finalized and combined all for a presentation-ready prototype for Jonathan

A. Construction Jenga Tower

B. Free-for-All Soccer

C. Maze and Mosaic

5. Challenges

No issues, but one of our biggest challenges has been balancing interesting gameplay with the project's technical and production limitations.

1. Supporting players who join and leave

- The game must work without requiring everyone to start or finish together.
- We need to determine what happens to a character when its player disconnects.
- We also need to decide whether individual scores persist, how players return, and whether their absence affects the remaining participants.
- Any solution must avoid interrupting the game for everyone else.

2. Keeping the rules simple

- Visitors should understand the objective without needing a long tutorial.
- Too many mechanics can make the game hard to learn, especially when several players are active at once.
- Our visual prototypes need to communicate what players can do, what they should aim for, and what happens after an action.

3. Making the screen readable with 20–30 players

4. Why would people want to play or stay

6. Where Are We Now?

- We had the meeting with the big boss
- Out of the three, he took the maze mosaic
- Through the meeting, we made some changes in the game

7. Where Are We Going?

- All the elements are in place; now we focus on two things

- The artistic visuals for the game
- The gameplay mechanics/testing and de-risking

Overall, we're focusing our efforts on the visuals and mechanics of the game for the upcoming